Assignment 3 - I2C: Temperature and humidity 100%

11.2.2024

Forsøk 1

Se gjennom tilbakemelding
10.2.2024

Forsøk 1 Poengsum:
100%

Vis tilbakemelding

Anonym vurdering: **nei**

Ubegrenset antall forsøk tillatt
16.12.2023 til 29.4.2024

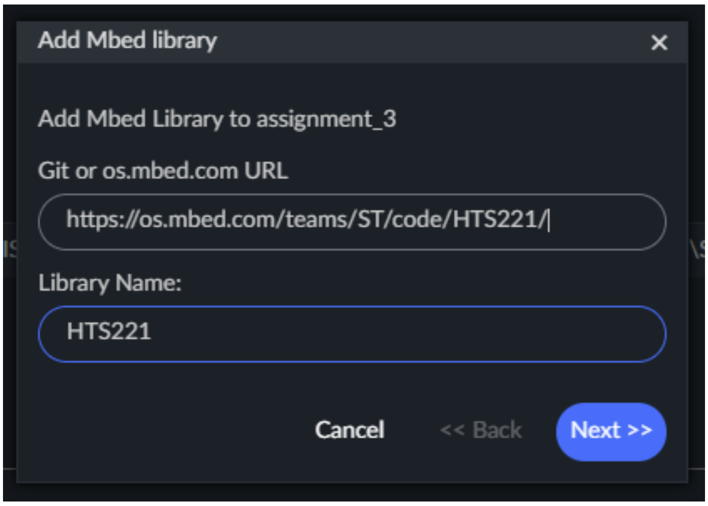
Detaljer

Preparations

This assignment assumes you have completed all steps of the previous assignments. The resources and general instructions from previous assignments are still useful here, this assignment will only list new relevant resources.

1. Create a new program for target DISCO-L475VG-IOT01A (B-L475E-IOT01A) under git folder (*<course folder>/assignments/*) called *assignment_3*. You can choose to create a new program based on example *mbed-os-example-blinky-baremetal* or simply take a copy of a previous assignment. In any cases make sure to link *mbed-os* library to the shared instance you previously created and make a clean build.

2. Install the STM32 HTS221 library to your project from URL [https://os.mbed.com/teams/ST/code/HTS221/ (https://os.mbed.com/teams/ST/code/HTS221/)](https://os.mbed.com/teams/ST/code/HTS221/)



3. Add Mbed OS version of the [DFRobot_RGBLCD1602 (https://uia.instructure.com/courses/14984/files/2471064?wrap=1)](https://uia.instructure.com/courses/14984/files/2471064?wrap=1) (https://uia.instructure.com/courses/14984/

files/2471064/download?download_frd=1) library. Extract the files next to your main.cpp file in Mbed Studio. The constructor takes a pointer to an I2C instance, addresses to RGB and LCD devices, number of columns and rows. All constructor parameters have default values, but as minimum you must give a pointer to an object of class DevI2C.

```
I2C lcdI2C(D14, D15);
DFRobot_RGBLCD1602 lcd(&lcdI2C); // Note: If you are using display V1.0, you will need to add RG
B_ADDRESS_V10_7BIT as second parameter
lcd.init(); lcd.diplay(); lcd.printf("Hello World!");
```

4. You will probably use float in printout, e.g. printf("Temp: %.1f C", temp). Open file mbed_app.json and set:

```
"platform.minimal-printf-enable-floating-point": true,
"platform.stdio-minimal-console-only": false
```

Resources

- **LCD examples (https://github.com/DFRobot/DFRobot_RGBLCD/tree/master/examples)** (Note: Arduino examples, but you will see examples of to use the DFRobot LCD Library)
- **HTS221 API description (https://os.mbed.com/teams/ST/code/HTS221/docs/tip/classHTS221Sensor.html)**
- **Board pinouts (https://os.mbed.com/platforms/ST-Discovery-L475E-IOT01A/)**
- **InterruptIn (https://os.mbed.com/docs/mbed-os/v6.16/apis/interruptin.html)**
- **Board documentation (https://www.st.com/resource/en/user_manual/dm00347848-discovery-kit-for-iot-node-multichannel-communication-with-stm32l4-stmicroelectronics.pdf)** (see page 10 for sensor location)

Requirements

In this assignment you will create a temperature / humidity device. Measured values comes from the HTS221 sensor that is mounted on the board connected to I2C2. These values are then displayed on the LCD which you must connect over I2C1. To test temperature put a finger on the temperature / humidity sensor to heat it up. Breathe on it to test humidity.

1. 5% Connect a button to a digital input.
2. 10% Connect the RGB LCD to I2C1 and VCC/GND (see underside of the LCD).
3. 10% Read the temperature and humidity values from the sensor and print them (use serial monitor to see them).
4. 20% Show the current temperature on the LCD. The value should update regularly (use 1, 2 or 3 seconds).
5. 25% Toggle which value is shown with the button. Default is temperature. Press once and it changes to humidity. Press and it changes back.
6. 15% Set the backlight color of the LCD based on the temperature when showing temperature. < 20 C: blue. >= 20 <= 24: orange. > 24: red.
7. 15% Set the backlight color of the LCD based on the humidity when showing humidity. 0% humidity: white. Then gradually fade to blue (how blue = humidity). 100%: Fully blue.

Delivery

Deliverer the following as text in Canvas:

- Link to Bitbucket where to code has been delivered.
  Directly to the file if there is only 1, or the directory if more than 1.
- Link to YouTube (or other video service) with video showing each of the requirements.
  Make sure that each requirement you have completed is clearly shown.
  You don't need to show the actual code, just which of the requirements are met.